

**Homework 4**

1. Suppose that operator  $T : B \rightarrow B$  defined on the set of bounded function  $B(R_+)$  is given by

$$TV(x) = \beta - \frac{\alpha}{x+1} + \delta V(x), \quad \delta \in (0, 1).$$

- (a) Using Blackwell Sufficient Condition and a Contraction Mapping Theorem, show that operator  $T$  has a unique fixed point
- (b) Find the fixed point of operator  $T$  using two different approaches: directly by solving the functional
  - (i) directly, i.e. by solving the functional equation  $V = TV$ ;
  - (ii) as a limit of  $T^n(V_0)$  where  $V_0 = 0$  (i.e., find  $V_1(x)$ ,  $V_2(x)$ , ...,  $V_n(x)$ , and take the limit as  $n \rightarrow \infty$ ).

2. **Optimal growth model with endogenous population growth**

Note: this problem was used in the 2025 prelim exam

Consider a variation of the optimal growth model with *endogenous* population growth and  $AK$  production technology. That is, in period  $t$  it produces output  $AK_t$ ,  $A > 0$  if it has aggregate capital  $K_t$ . The aggregate output can be split between aggregate consumption  $C_t$  and aggregate capital  $K_{t+1}$  saved for the next period's production. That is, the feasibility constraint is

$$C_t + K_{t+1} = AK_t.$$

The aggregate consumption is split equally among  $N_t$  people in the economy in each period. The number of people grows endogenously, depending on per capita consumption, according to the law of motion:

$$N_{t+1} = g \cdot N_t \cdot \left( \frac{C_t}{N_t} \right)^\gamma = g \cdot N_t^{1-\gamma} C_t^\gamma, \quad \gamma \in (0, 1), \quad g > 0 \quad (1)$$

The planner's objective is to maximize the discounted flow of per capita utilities:

$$\sum_{t=0}^{\infty} \beta^t \log \left( \frac{C_t}{N_t} \right), \quad \beta \in (0, 1).$$

The economy's capital stock and population in period 0 are  $K_0$  and  $N_0$ , respectively.

- (a) Re-write the feasibility constraint in per capita terms, using  $c_t = C_t/N_t$  and  $k_t = K_t/N_t$  for per capita consumption and capital, respectively.
- (b) Set up the planner's problem recursively (Hint: it might be convenient to use per capita consumption as control variable).
- (c) Using the guess-and-verify method, find the optimal policy for per capita consumption.
- (d) Describe how per capita consumption evolves over time depending on the economy's initial conditions  $(K_0, N_0)$ .

- (e) Your classmate states that the law of motion for population (1) implies that as  $g$  increases, long-run population growth rate rises. Is your classmate correct? Justify your answer. If you think that your classmate is wrong, analyze how long run population growth rate is impacted by  $g$  and explain your findings intuitively.

### 3. Comparative statics in the optimal growth model

Consider an optimal growth model with general  $u(c)$  and  $f(k)$ . Suppose also that, in addition to capital income  $f(k)$ , agents receive a fixed income  $y > 0$  in every period. How does an increase in  $y$  affect the optimal policy functions for consumption  $c(k)$  and capital  $k'(k)$ ? (Hint: you might need to use a recursive argument to understand how a change in  $y$  affect the slope of  $V(k)$ . Establish the effect on  $c(k)$  first, and then think about the effect on  $k'(k)$ .)

### 4. Skill accumulation problem

Note: this problem was used in the 2021 prelim exam

Consider a discrete-time infinite horizon decision problem of an individual who decides how much to invest in skill accumulation in every period. If in period  $t$  the individual has skill  $s_t$ , she supplies  $s_t$  units of effective labor to the labor market, for which she is compensated at rate  $w$  (constant across periods). The labor income can be spent to finance consumption  $c_t$  in period  $t$  or to invest  $i_t$  in skill accumulation. The law of motion for the individual's skill is given by

$$s_{t+1} = (1 - \delta)s_t + As_t^{1-\gamma}i_t^\gamma, \quad t \geq 0,$$

where  $\delta, \gamma \in (0, 1)$  and  $A > 0$ . The individual's life-time utility is given by

$$\sum_{t=0}^{\infty} \beta^t u(c_t),$$

where  $\beta \in (0, 1)$  and  $u : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  is a strictly increasing, strictly concave, and continuously-differentiable function, such that  $\lim_{c \rightarrow 0} u'(c) = +\infty$ .

No storage technology is available, and the individual has no access to asset markets.

- Set up the individual's decision problem recursively. Derive the Euler equation and interpret it intuitively.
- From now onwards, assume that  $u(c) = \ln(c)$ . Using a "guess and verify" approach, show that the individual's optimal investment choice is linear in her skill.
- How does the individual's optimal investment in skill vary with  $\delta$ ? Provide a formal argument and an intuitive interpretation.